Boundary value problems

We're already very familiar with initial value problems. Boundary value problems are similar.

Boundary conditions and boundary value problems

Boundary value problems let us use boundary conditions that are specified for different values of the independent variable. Contrast this with initial value problems, in which we use initial conditions to define the value of the function and its derivatives all for the same value of the independent variable.

For instance, we would typically use the following set of initial conditions to solve a second order ordinary differential equation.

$$y(t_0) = y_0$$

$$y'(t_0) = y_0'$$

We can tell that this is a pair of initial conditions because they define y and its derivative y' at the same value t_0 of the independent variable.

On the other hand, for the same second order ordinary differential equation, we could use any of these pairs of boundary conditions.

$$y(t_0) = y_0 \qquad y(t_0) = y_0 \qquad y'(t_0) = y_0 \qquad y'(t_0) = y_0$$

$$y(t_1) = y_1 \qquad y'(t_1) = y_1 \qquad y(t_1) = y_1 \qquad y'(t_1) = y_1$$

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The first pair defines two values of y, the last pair defines two values of y', and the other two pairs each define one value of y and one value of y'. But every pair defines values of y and/or y' at two different values of the independent variable, t_0 and t_1.

No matter which pair of boundary conditions we have, our process for solving boundary value problems will look just like the process we're used to for solving initial value problems.

Keep in mind though that, unlike initial value problems, with boundary value problems it's common to find no solutions, one solution, or many solutions. But we care about boundary value problems because we use them all the time with partial differential equations to model real-world phenomena.

Boundary conditions for the product solution

With that in mind, we want to go back to the separation of variables process we were looking at earlier.

We'd been working with the partial differential equation,

$$\frac{\partial u}{\partial t} = k\frac{\partial^2 u}{\partial x^2}$$

and we used the product solution $u(x,t) = v(x)w(t)$ to rewrite the PDE as an ODE instead,

$$v(x)\frac{dw}{dt} = k\frac{d^2v}{dx^2}w(t)$$

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we separated variables,

$$\left(\frac{1}{kw(t)}\right)\frac{dw}{dt} = \left(\frac{1}{v(x)}\right)\frac{d^2v}{dx^2}$$

$$\left(\frac{1}{kw(t)}\right)\frac{dw}{dt} = \left(\frac{1}{v(x)}\right)\frac{d^2v}{dx^2} = -\lambda$$

and then split the equation in two, putting both in terms of the separation constant $-\lambda$. The result is a pair of ordinary differential equations, one first order equation and one second order equation, and we'll look later at how to solve these.

$$\frac{dw}{dt} = -k\lambda w(t)$$

$$\frac{d^2v}{dx^2} = -\lambda v(x)$$

But now let's consider what happens when we have the boundary conditions $u(0,t) = 0$ and $u(L,t) = 0$ to accompany the partial differential equation.

This partial differential equation is an equation for u defined in terms of the independent variables x and t, which is why the solution we find will be $u(x,t)$. We think about x as the spatial variable and t as the partial variable. That's why our boundary conditions $u(0,t) = 0$ and $u(L,t) = 0$ are defined for different values of x; we're defining the function's value at the spatial boundary $x = 0$ and $x = L$. If we were given any initial conditions, they would be defined for different values of t, the time variable.

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We're using the product solution $u(x,t) = v(x)w(t)$. If we substitute the first boundary condition, we get

$$u(0,t) = v(0)w(t) = 0$$

Given the product $v(0)w(t) = 0$, we know either $v(0) = 0$, or $w(t) = 0$. But if $w(t) = 0$, we'll just get the trivial solution $u(x,t) = 0$. Only $v(0) = 0$ will give us an interesting solution.

And if we substitute the second boundary condition, we get

$$u(L,t) = v(L)w(t) = 0$$

Given the product $v(L)w(t) = 0$, we know either $v(L) = 0$, or $w(t) = 0$. But if $w(t) = 0$, we'll again just get the trivial solution $u(x,t) = 0$. Only $v(L) = 0$ will give us an interesting solution.

So the partial differential equation boundary value problem

$$\frac{\partial u}{\partial t} = k\frac{\partial^2 u}{\partial x^2}$$

with $u(0,t) = 0$ and $u(L,t) = 0$

has at this point turned into two separate problems, one first order equation and one second order equation, with two boundary conditions.

$$\frac{dw}{dt} = -k\lambda w(t)$$

$$\frac{d^2v}{dx^2} = -\lambda v(x)$$

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with $v(0) = 0$ and $v(L) = 0$

Later on we'll look at how to solve this particular boundary value problem, but right now let's just do a simple boundary value problem so that we understand how to apply boundary conditions to find the actual solution from the general solution.

Example

Solve the boundary value problem, if $y(0) = 0$ and $y(2) = -1$.

$$y'' - 2y = 0$$

The associated characteristic equation is

$$r^2 - 2 = 0$$

$$r^2 = 2$$

$$r = \pm\sqrt{2}$$

which means the general solution to the homogeneous equation is

$$y(t) = c_1 e^{\sqrt{2}t} + c_2 e^{-\sqrt{2}t}$$

Substituting $y(0) = 0$ into the general solution gives

$$0 = c_1 e^{\sqrt{2}(0)} + c_2 e^{-\sqrt{2}(0)}$$

$$0 = c_1 + c_2$$

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$$c_1 = -c_2$$

Substituting $y(2) = -1$ into the general solution gives

$$-1 = c_1 e^{\sqrt{2}(2)} + c_2 e^{-\sqrt{2}(2)}$$

$$-1 = c_1 e^{2\sqrt{2}} + c_2 e^{-2\sqrt{2}}$$

$$-1 = -c_2 e^{2\sqrt{2}} + c_2 e^{-2\sqrt{2}}$$

$$-1 = c_2(e^{-2\sqrt{2}} - e^{2\sqrt{2}})$$

$$c_2 = -\frac{1}{e^{-2\sqrt{2}} - e^{2\sqrt{2}}}$$

So we get

$$c_2 = \frac{1}{e^{2\sqrt{2}} - e^{-2\sqrt{2}}}$$

$$c_1 = \frac{1}{e^{-2\sqrt{2}} - e^{2\sqrt{2}}}$$

Then the solution to the boundary value problem is

$$y(t) = \frac{e^{\sqrt{2}t}}{e^{-2\sqrt{2}} - e^{2\sqrt{2}}} + \frac{e^{-\sqrt{2}t}}{e^{2\sqrt{2}} - e^{-2\sqrt{2}}}$$

So just like with initial value problems, solving boundary value problems will require us to find the general solution first, then substitute boundary

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conditions into the general solution and/or its derivatives in order to find values of the constants c_1 and c_2.

And remember that, even though we found a solution to the boundary value problem in the last example, it's common to see boundary value problems with no solution, when there's no solution to the system of equations we create for c_1 and c_2 after substituting the boundary conditions.

It's also common to see boundary value problems with many solutions, when the system of equations we create for c_1 and c_2 after substituting the boundary conditions gives a value for one constant that relies on the value of the other. For instance, when we find that c_2 can be any value, and c_1 is only defined as $c_1 = 2c_2$.

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